HMC ONE
AI STRATEGY. BUILT TO SCALE.

CLIENT CASE STUDY · BEVERAGES · SENSORY & INNOVATION

SPIRITS SENSORY SCORECARD · FRUIT · WOOD · SPICE · SMOKE

# Flavour Intelligence

Sensory scores became a shared flavour story: compare whole profiles, see what makes a product distinctive, and put price beside the attribute that matters.

1 → 12 from a single average to sensory dimensions on one fixed scale



**3** views
Original dashboard views
radar · price · 1-v-1



**19** attributes
In the reference scorecard
a shared sensory vocabulary



**8** prototypes
Profiles in the interactive lab
synthetic, coded samples

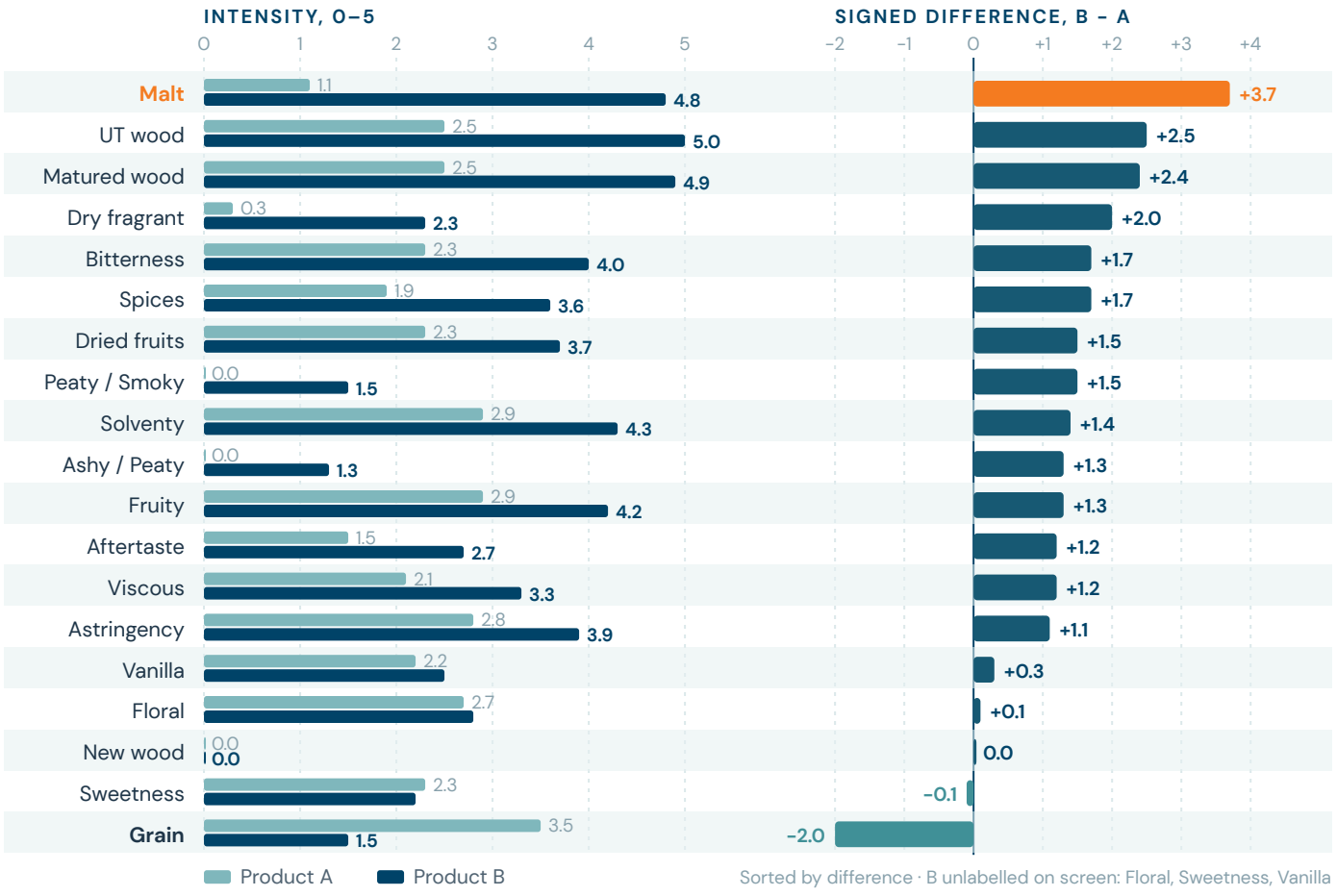


**12** dimensions
In the live explorer
linked across every visual

Rings: one segment per item: the three original views, the 19 reference attributes, the 8 lab prototypes (orange: the balanced candidate closest to target) and the 12 explorer dimensions.

## 01 Two products, nineteen attributes, one clear gap

Redrawn from the supplied 1-v-1 dashboard screen · product names coded




Sorted by difference · B unlabelled on screen: Floral, Sweetness, Vanilla

**LARGEST GAP**
**Malt: +3.7**
Product B scores 4.8 against A's 1.1. No average would show a gap this size.

**THE REVERSAL**
**Grain runs the other way**
A is stronger on grain (3.5 vs 1.5, -2.0). Sweetness is the only other attribute where A leads, by 0.1.

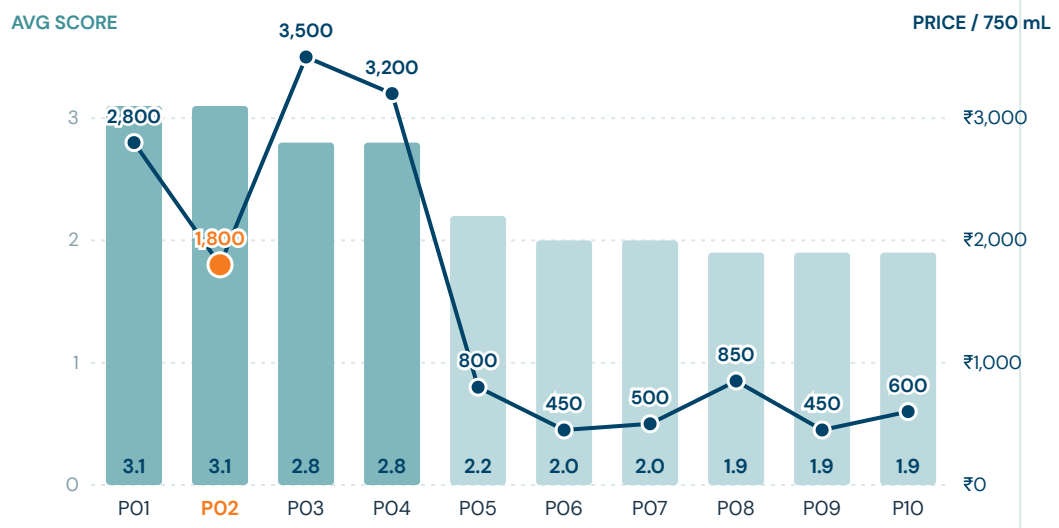
**WOOD CARRIES B**
**UT wood +2.5, matured wood +2.4**
B sits at 5.0 and 4.9: the top of the scale on both wood attributes.

**HOW TO READ IT**
**Difference, not improvement**
A positive bar means more intensity in B. It is not a quality or preference score.

## 02 Price beside the profile, shape beside the average

Left: supplied value-vs-price screen · right: illustrative lab data

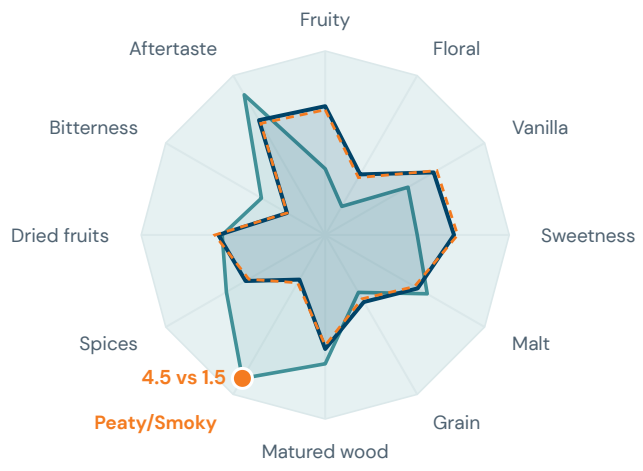
**Average sensory score and price, 10 products** — REFERENCE SCREEN
Coded P01–P10, ordered by average score




Bars: average sensory score · line: reported price (one city) · Price filter on the screen: ₹450–3,500

**P02** matches the top average (3.1) at ₹1,800, about half of P03's ₹3,500. Price only becomes a useful conversation once the sensory side sits next to it.

**Similar averages, very different fingerprints** — ILLUSTRATIVE DATA
Synthetic prototypes, fixed 0–5 scale, synthetic target




MEAN INTENSITY
2.67 S-06 · 2.76 S-04

TARGET FIT / 100
98.1 S-06 · 76.3 S-04

KEY
S-06 Balanced candidate
S-04 Smoky depth
Target (dashed)

Averages 0.09 apart; the smoke axis alone separates them. Fit = 100 × (1 − RMSE ÷ 5), a descriptive similarity index.

## 03 Four pain points, four fixes

The intended workflow shift, without assuming a measured time saving

BEFORE
**Hidden fingerprints**
Similar averages hid very different mixes of fruit, malt, wood, spice and smoke.
RADAR ON A FIXED SCALE
AFTER
**The whole shape**
Every profile read as a complete sensory fingerprint on the same 0–5 scale.

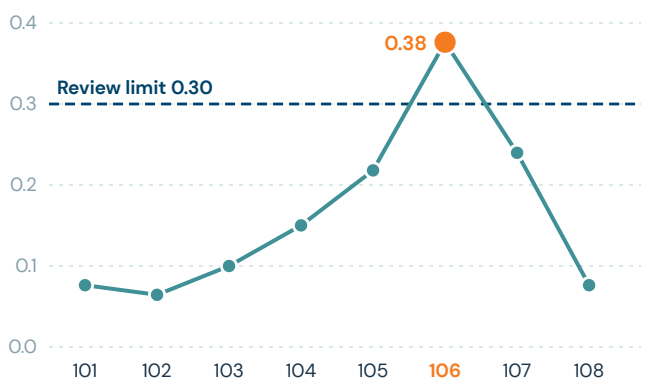
BEFORE
**Complex comparisons**
Reviewers compared columns and calculated gaps in their heads.
PAIRED BARS · SIGNED GAPS
AFTER
**Differences explicit**
Paired attribute intensities with the signed difference beside each one.

BEFORE
**Price without context**
Price was reviewed separately from the sensory observations.
ATTRIBUTE BESIDE PRICE
AFTER
**Price in the same view**
A chosen sensory attribute placed directly beside price.

BEFORE
**Separate scorecards**
Static screenshots handed on without the underlying context.
ONE SENSORY DICTIONARY
AFTER
**One shared language**
Consistent descriptors and scales; scores exported with dataset label and comparison.

## 04 When a batch drifts

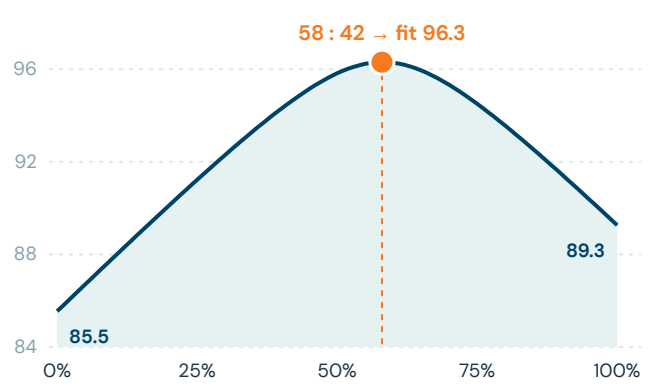
Eight demo batches against the target · illustrative rule




Batch B-1xx · RMSE vs target across 12 dimensions
B-106 flagged: matured wood +0.7 (limit 0.50)

## 05 When two profiles meet

Target fit of a linear mix · illustrative, confirm by tasting




Share of S-01 Orchard lift in a mix with S-03 Oak structure
Weighted scenario price at 58 : 42 = ₹3,040 · RMSE 0.185

## 06 How HMC ran it

1 Discover — **Define the questions**
Which profiles are similar, which attributes explain a gap, where price fits. → Question map.

2 Design — **Align the scorecard**
Rating anchors, fixed scale, intensity kept apart from preference. → Sensory dictionary.

3 Build — **Connect the views**
Radar, paired bars and linked filters on one selection. → Connected dashboard.

4 Adopt — **Guide the review**
Choose, inspect the strongest gaps, record the next tasting question.

5 Scale — **Extend the questions**
3D flavour map, blend scenarios and batch review, assumptions visible.

## What a sensory team can take forward

**01 Compare the shape**
The whole profile explains more than a single average. Keep the attribute vocabulary and scale consistent.

**02 Make differences explicit**
Pair every overview with an attribute-level comparison so reviewers can explain what changed.

**03 Keep the judgement human**
Use interactive scenarios to focus the next tasting and discussion. Validate decisions with the sensory team.

Built with **Power BI** · **D3.js** · **Three.js / WebGL** · **JavaScript** · **CSV export** | Prepared by Havish M Consulting · Case-study reconstruction from supplied screens, product names coded. Lab profiles, prices, batches and blend outputs are synthetic illustrations.

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